Paper OOL-3: Magnetite Networks and Distributed Electron Transport: Mixed-Valence Coupling Before Biology

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Abstract

This paper separates distributed electron transport from the local Fe–S catalysis of Paper 2. Magnetite-like and mixed-valence mineral networks are treated as possible redox couplers, not as independent origins of life. The key question is whether a connected mineral graph can move reducing equivalents between separated donor, acceptor, and reaction zones. The companion toy model supports the narrow claim of adaptive conductance improvement while leaving real mineral percolation, defect chemistry, and endpoint product tests open.

Symbol Conventions

CDFD origins-of-life symbol convention.

Symbol	Meaning in Part II	Origins-of-life reading
Φ	Available driving flux	Redox, thermal, photochemical, proton/ion, or chemical potential drive supplied by the environment
C	Effective constraint or resistance	Mineral geometry, pore confinement, diffusion barrier, permeability, activation barrier, or maintenance burden
S	Responsive organization	Surface, boundary, catalytic network, coacervate, or protocellular structure that routes flux into retained function
M_s	Retained structural memory	Composition, sequence, topology, compartment state, mineral imprint, or boundary history that persists across renewal cycles
Ψ_s	Local adaptive ratio $(\Phi/C) \cdot S \cdot M_s$	Bookkeeping gauge for whether drive, constraint, responsiveness, and retention are co-present; not a universal constant
Λ_{life}	Integrated Life Number where used	Capstone surplus ratio for decay, near-critical proto-biology, and sustained self-maintenance

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Accumulation or reinforcement coefficient	Defines how fast a constraint, pathway, or retained structure grows under load
β	Relaxation, decay, or washout coefficient	Defines recovery, loss, clearance, hydrolysis, or return toward baseline
γ	Coupling, diffusion, or redistribution coefficient	Defines spread across pores, surfaces, compartments, networks, or chemical phases
δ, ϵ	Perturbation, tolerance, or small offset scales	Used only as local thresholds, numerical guards, or experimental tolerances
η, λ, μ, ν	Efficiency, rate, baseline, or frequency terms	Meaning is set by the equation and subscript where they occur
ρ, σ, τ, ϕ	Density/correlation, conductance/noise, time-scale, or phase/field terms	Never universal constants unless a paper explicitly defines them that way
Ω, Λ	Domain/state space; Life Number where used	Ω names a modeled state space; Λ is reserved for life-number notation only where defined

1 Introduction

Metabolism, even in protocellular form, requires spatial integration: donors and acceptors are not co-located. Minerals that only catalyze local reactions cannot, by themselves, span separation scales. Magnetite's inverse spinel structure hosts Fe^{2+} and Fe^{3+} on octahedral sites, enabling thermally activated electron hopping [4]. In the OOL narrative, magnetite-like phases are therefore candidates for **distributed** redox coupling atop Fe–S surface chemistry [2, 3].

Unresolved gap carried from Paper 2: Fe–S establishes interfaces; Paper OOL-3 asks whether *extended* conduction can self-organize transport bottlenecks into usable pathways.

Status: Inferred

Capstone channel mapping. Mixed-valence magnetite implements the sketch's *electron-motion* strategy ($Fe^{2+} \leftrightarrow Fe^{3+}$ hopping). In Paper 12 notation this maps to the conductance proxy σ_e appearing in $J = I_E \sigma_e \sigma_p / B_{\text{stab}}$; adaptive erosion or channelization of C is the discrete-network caricature of raising C_{electron} or σ_e until another capacity becomes limiting.

2 Model and Assumptions

2.1 Notation

We retain $\Psi_s = (\Phi/C) \cdot S \cdot M_s$ for the local equilibrium ratio. The constraint field C is nonnegative; small C means "easy" transport in the phenomenological flux law

$$\boldsymbol{J}_e = -\frac{1}{C}\nabla\Phi. \tag{1}$$

When C changes over time, we read it as annealing, dissolution, reprecipitation, or, in the toy code, erosion rules. Those rules lower resistance along high-flux corridors.

Status: Derived

2.2 Crystal chemistry and hopping

Electron transfer between adjacent iron sites,

$$\mathrm{Fe}^{2+} \leftrightarrow \mathrm{Fe}^{3+} + e^-, \tag{2}$$

supports nonmetallic conduction with thermally activated rates $k \propto \exp(-E_a/k_B T)$. Unlike free-electron metals, transport is localized and structure-sensitive; defects and grain boundaries dominate macroscopic resistance.

Status: Inferred

2.3 Network geometry

In rocks, magnetite often appears as anastomosing grains rather than isolated cubes. Intergrowth topology matters: percolation thresholds control whether a conductive cluster spans the sample. We do not compute percolation explicitly here; the numerics use a 2D grid with two fixed-potential "poles" and allow C to adapt.

Status: Hypothesis

2.4 Toy adaptive-constraint dynamics (script-level)

The supplementary code combines diffusive relaxation of Φ with rules that reduce C where $|\nabla(\Phi/C)|$ is large. This is a stand-in for "flow opens channels" feedback. It is *not* a first-principles model of magnetite defect chemistry.

Status: Hypothesis

3 Main Results

3.1 Physical narrative

At field level, magnetite couples distant Fe–S reaction sites: electrons may traverse the oxide lattice or grain-contact networks, while protons remain chiefly aqueous (OOL-3). Magnetic order, spin-dependent hopping, and defect chemistry add structure beyond a scalar C field; these effects are noted but not resolved.

Status: Inferred

3.2 Constraint feedback (qualitative)

Motivated by adaptive flux limitation (AFL) style laws used elsewhere in the CDFD/AFL manuscripts, one writes schematically

$$\frac{\partial C}{\partial t} = \alpha|\boldsymbol{J}_e| - \beta C + \text{diffusion}, \tag{3}$$

signifying that sustained current can anneal pathways (lower C) while aging or precipitation raises resistance. Coefficients are environment-specific; we do not fit them to experiments in this manuscript.

Status: Hypothesis

4 Numerical Verification

Run `supplementary_ool_03.py`. The script reports "network efficiency" from boundary flux metrics under adaptive C.

Observed outputs:

- Initial network efficiency: 0.500
- Final network efficiency: 9.894
- Minimum final constraint: 0.100
- Maximum final constraint: 0.847

The dramatic efficiency increase supports the *model* claim that erosion-style feedback can lower global resistance. The narrow final range of C suggests near-uniform lowering in this parameter set; localized "wire" formation is not clearly resolved. Claims should emphasize global optimization unless future metrics (e.g., conductance centralities) show channelization.

Status: Derived

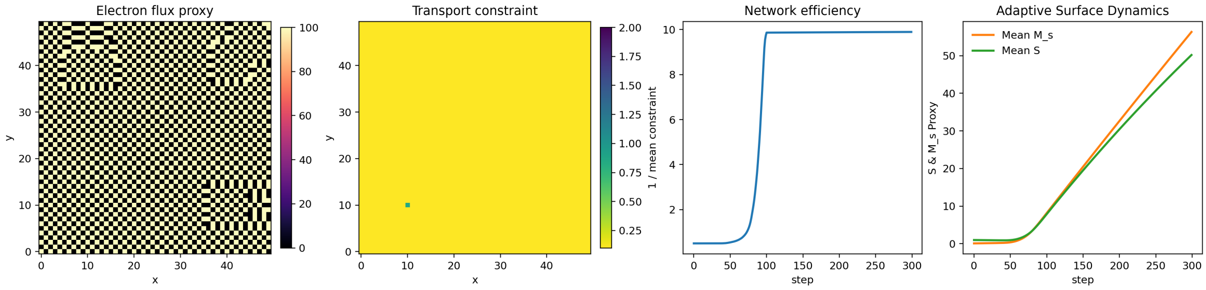

Figure 1: Magnetite-like transport diagnostic. The toy model lowers transport resistance under sustained drive, but the final constraint field is nearly uniform in this parameterization. The figure therefore supports distributed conductance improvement, not a strong claim of sharply localized mineral wires.

5 Why Magnetite Belongs After Fe–S

Fe–S interfaces supply localized catalytic nodes. Magnetite-like phases add a different function: distributed electron transport across a larger mineral domain. The logical transition is:

$$\text{localized redox node} \rightarrow \text{mixed-valence transport network.} \tag{4}$$

This matters because early reaction zones need both local activation and long-range balancing. A single pore wall can reduce barriers, but a connected oxide/sulfide network can move reducing equivalents between sites that are not chemically identical.

Status: Inferred

6 Network Diagnostics Missing from the Toy Model

The current script reports mean resistance and a final field. A stronger future version should measure:

- source–sink conductance rather than global mean constraint;
- percolation probability across many random mineral geometries;
- persistence of corridors after forcing is removed;
- sensitivity to dissolution, precipitation, and sulfidation;
- whether electron transport co-localizes with proton/ion coupling from Paper 4.

These diagnostics would separate true channel formation from the near-uniform erosion observed here.

Status: Open

7 Percolation, Defects, and Mineral Mosaics

Magnetite is not merely an oxide label; its relevance here is the possibility of spatially extended, mixed-valence conduction. In a natural setting, however, the relevant object is rarely a pure magnetite block. It is a mineral mosaic containing oxides, sulfides, carbonates, silica, clays, grain boundaries, fractures, inclusions, and redox-altered rims. The transport question is therefore topological:

$$P_{\text{span}} = P(\mathcal{G}_{\text{mv}} \text{ connects donor to acceptor})\,, \tag{5}$$

where $\mathcal{G}_{\text{mv}}$ is the graph of mixed-valence or otherwise redox-conductive contacts. A few highly conductive grains do not matter if they are isolated. Conversely, a moderately conductive but spanning network can matter because it couples separated reaction zones.

Defects also cut both ways. Vacancies, impurities, and grain contacts can increase hopping sites, create trap states, or interrupt conduction. In the coarse CDFD notation these effects are bundled into $C_e(\boldsymbol{x})$, but the physical interpretation should remain explicit:

$$C_e(\boldsymbol{x}) = C_{\text{lattice}} + C_{\text{grain}} + C_{\text{defect}} + C_{\text{fluid/contact}}. \tag{6}$$

The paper's strong claim is not that magnetite always lowers C_e. The defensible claim is conditional: if a mixed-valence phase forms a spanning redox graph and terminates at chemically active interfaces, then it can reduce effective source–sink resistance relative to disconnected mineral grains.

Status: Inferred/Open

8 Endpoint Chemistry Is the Real Test

Electrical or electrochemical conductance alone is not enough. The origin-of-life claim only becomes meaningful if a conductive network changes chemistry at its endpoints. A clean experimental score would compare product yields with and without a connected mixed-valence bridge:

$$\Delta Y_{\text{endpoint}} = Y_{\text{connected}} - Y_{\text{disconnected}}, \tag{7}$$

under matched donor, acceptor, pH, temperature, mineral mass, and surface-area conditions. The network claim is supported when $\Delta Y_{\text{endpoint}} > 0$ for products requiring transported reducing equivalents, not merely when an electrode detects current.

This distinction prevents an overclaim. Magnetite-like transport is a *coupler*. It should alter downstream product distribution, redox state, or persistence time at a reaction interface. If it only dissipates energy as heat or background current, it has not advanced the OOL sequence.

Status: Derived experimental guardrail

9 Transport Networks as Couplers, Not Origins by Themselves

Magnetite-like phases should not be presented as a second independent origin mechanism. Their role is to couple locations. A redox-active mineral network matters only if it connects an electron donor region, an electron acceptor region, and a reaction interface where chemical work can occur. The minimal coupling graph is:

$$\text{donor} \xrightarrow{\sigma_e} \text{transport network} \xrightarrow{\sigma_e} \text{reaction interface}. \tag{8}$$

If the network conducts but does not terminate in chemistry, it is just a dissipative wire. If the chemistry exists but the network is disconnected, it is just a local catalyst. Paper 2 is therefore the bridge between Paper 1's local Fe–S activation and Paper 4's proton/ion co-channel.

Status: Derived

10 Environmental Robustness Tests

Future magnetite-network claims should be tested across temperature, pH, ionic strength, sulfide activity, and wetting state. In a vent reactor, mineral membranes can grow, rupture, recrystallize, and change conductivity as fluids mix [1]. In the field, magnetite may coexist with sulfides, carbonates, clays, and silica phases, each modifying transport. CDFD therefore treats magnetite as an exemplar of mixed-valence electron distribution, not as a mandatory universal material.

The immediate measurable prediction is modest: adding a connected mixed-valence phase to otherwise similar mineral compartments should increase source–sink redox coupling and alter product distribution at downstream catalytic sites. That is the right experimental target, not an unsupported claim that any magnetite-bearing rock becomes proto-biological.

Status: Open

11 Interpretation

Magnetite provides a geologically plausible bridge from localized catalysis to spatially extended redox coupling [4]. The simulation strengthens the *conceptual* point—adaptive transport resistance can reorganize under forcing—without proving Archean network morphologies.

Status: Inferred

12 Limitations and Open Problems

- Real magnetite conduction is tensorial, frequency-dependent, and sensitive to oxyexsolution.
- Channel vs uniform erosion must be distinguished with topological summaries.
- Coupling to fluid flow and sulfidation pathways is absent.

Status: Open

13 Conclusion

Paper OOL-3 positions mixed-valence oxide networks as the distributed-transport counterpart to Fe–S interface catalysis. Numerics show strong efficiency gains in an idealized adaptive-resistance model; morphology claims remain tempered. OOL-3 introduces structured water and proton fields required for chemiosmotic coupling.

Status: Inferred

14 Reproducibility Note

```
python supplementary_ool_03.py
```

The script writes compact reproducibility outputs under `outputs/paper_02/`.

References

[1] Laura M. Barge et al. An origin-of-life reactor to simulate alkaline hydrothermal vents. *Journal of Molecular Evolution*, 79:213–227, 2014.

[2] William Martin and Michael J. Russell. On the origin of biochemistry at an alkaline hydrothermal vent. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 362(1486):1887–1925, 2007.

[3] Michael J. Russell, Roy M. Daniel, Allan J. Hall, and John A. Sherringham. A hydrothermally precipitated catalytic iron sulphide membrane as a first step toward life. *Journal of Molecular Evolution*, 39:231–243, 1994.

[4] Frances N. Skomurski, Sebastien Kerisit, and Kevin M. Rosso. Structure, charge distribution, and electron hopping dynamics in magnetite (Fe3O4) (100) surfaces from first principles. *Geochimica et Cosmochimica Acta*, 74(14):4234–4248, 2010.